

Emergent Cooperative Strategies for Pursuit-Evasion in Cluttered Environments: A Knowledge-Enhanced Multi-Agent Deep Reinforcement Learning Approach

Yihao Sun, Chao Yan, Han Zhou, Xiaojia Xiang and Jie Jiang

Abstract—Deep reinforcement learning (DRL) has recently emerged as a promising tool for tackling pursuit-evasion tasks. However, most existing DRL-based pursuit approaches still rely on individual rewards and struggle with complex scenarios. To address these challenges, we propose a knowledge-enhanced DRL approach for multi-agent pursuit-evasion in complex environments. Specifically, the cooperative pursuit problem is modeled as a decentralized partially observable Markov decision process from each pursuers perspective, where the team reward function is elaborately designed to encourage collaborative behavior and enhance team coordination. Then, a novel knowledge enhanced multi-agent twin delayed deep deterministic policy gradient (KE-MATD3) algorithm is presented to efficiently learn the cooperative pursuit policy. By integrating a knowledge enhancement mechanism that extracts effective information from an improved artificial potential field method, the cooperative pursuit policy achieves more robust convergence, mitigating the local optima that typically arise from individual reward-based learning. Finally, extensive numerical simulations and real-world experiments validate the efficiency and superiority of the proposed approach, demonstrating emergent cooperative behaviors among the pursuers.

I. INTRODUCTION

In recent years, multi-agent systems (MASs) have demonstrated significant potential in complex and dangerous tasks such as air combat, target interception, and urban air mobility management [1], [2], [3]. These applications typically involve multi-agent cooperative pursuit problems. The objective is to enable each pursuing agent to collaborate seamlessly with teammates while avoiding conflicts and adapting to dynamic environments, thereby ensuring that the target can be captured effectively [4].

To enhance the autonomy and adaptability of multi-agent systems in cooperative pursuit tasks, researchers have devoted significant efforts [5], [6]. Traditional methods, including Hamilton equation solutions [7], model predictive control

(MPC) [8], and feedback control [9], often require accurate kinematic models of agents and the environment. However, these models are challenging to obtain in practical scenarios.

To address this limitation, model-free deep reinforcement learning (DRL) [10], [11], [12] has gained increasing attention as a promising alternative. Various DRL-based approaches have been proposed to solve cooperative pursuit problems. For instance, Souza et al. [13] employed curriculum learning to progressively scale the pursuit policy from simpler to more complex tasks, but did not consider obstacles. Zhang et al. [14] introduced a CBC-TP Net framework for multi-UAV pursuit in urban settings, yet circumvented head-on collision issues by assigning different flight altitudes. Li et al. [15] adopted a modified MAPPO algorithm for multi-USV cooperative pursuit with obstacles, although their settings were constrained to sparsely distributed and fixed obstacle placements. Likewise, Xu et al. [16] proposed the MMOTS solution for underwater multi-AUV pursuit-evasion, though they assumed full state observability among pursuit AUVs and overlooked potential perception constraints. Overall, many existing approaches rely on simplified assumptions that fail to capture real-world complexities.

Furthermore, most current DRL-based pursuit approaches employ individual reward settings [17], [18], [19], [20], [21], which typically lead agents to prioritize their individual interests. Approaches could be particularly problematic in collaborative tasks requiring team coordination, potentially resulting in suboptimal behavior patterns. For instance, in traffic management systems, rewarding individual intersection efficiency might exacerbate congestion at adjacent junctions [22]. In contrast, team rewards promote collective goal consideration, motivating agents to maximize returns from a team perspective. However, even state-of-the-art RL algorithms such as MATD3 [23] and MAPPO [24] struggle to perform effectively when dealing with team rewards due to the notorious credit assignment problem [25]. While these algorithms may show some success in simple tasks using team rewards, such as multi-agent particle environments (MPE) [26], they often falter as task complexity increases.

To foster collaborative behaviors among perceptually limited pursuers in complex environments, this paper proposes a team reward based knowledge-enhanced multi-agent twin delayed deep deterministic policy gradient (KE-MATD3) algorithm. The main contributions are summarized as follows:

- A team reward based MADRL approach is proposed

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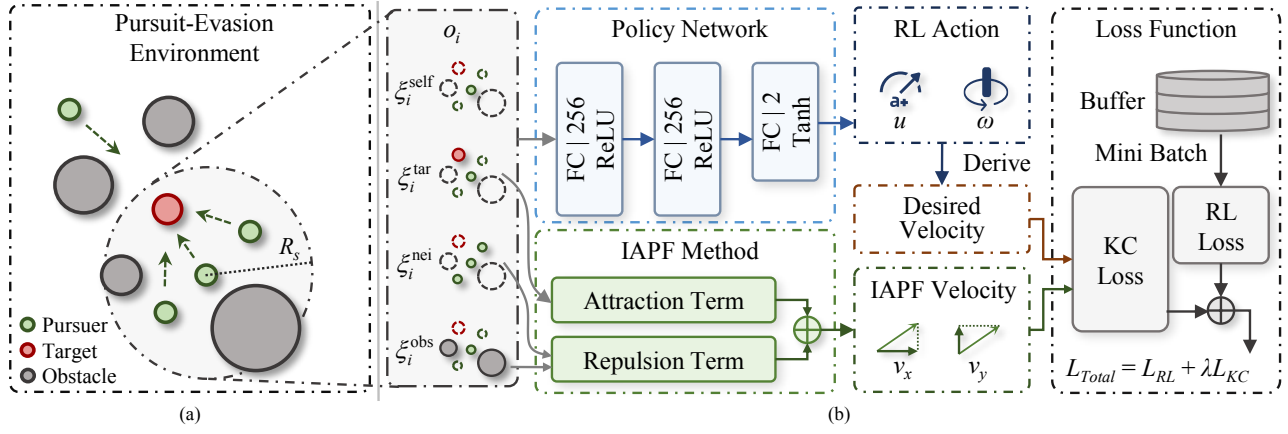


Fig. 1. The system framework of the cooperative pursuit task. (a) The multi-agent pursuit-evasion environment. (b) The proposed KE-MATD3 algorithm.

for multi-agent cooperative pursuit in cluttered environments, where the task is modeled as a decentralized partially observable Markov decision process.

- A knowledge enhanced (KE) mechanism is introduced to leverage insights from an improved artificial potential field method (IAPF), thereby facilitating the learning of challenging team rewards.
- The emergence of cooperative behaviors among pursuers is verified through both simulation and physical experiments.

II. PROBLEM FORMULATION

In this section, we model the multi-agent cooperative pursuit problem as a decentralized partially observable Markov decision process (Dec-POMDP).

A. Problem Description

As illustrated in Fig. 1(a), we consider a dynamic multi-agent pursuit-evasion system consisting of a moving target and a set of N homogeneous pursuers $\mathcal{P} := \{1, 2, \dots, N\}$. All agents move within a bounded domain containing obstacles. The obstacle set is defined as $\mathcal{O} := \{1, 2, \dots, M\}$, where M is the number of obstacles.

Let $p_i = (x_i, y_i)$ and v_i denote the position and velocity of pursuer i respectively, and $p_{tar} = (x_{tar}, y_{tar})$ represent the target's position. Each obstacle $j \in \mathcal{O}$ is characterized by its position $p_j^{obs} = (x_j^{obs}, y_j^{obs})$ and radius r_j^{obs} . Pursuers can only perceive neighboring pursuers and obstacles within a sensing range R_c . The target moves according to an artificial potential field method, with a maximum speed 1.5 times the pursuers' maximum speed v_{max} . The objective of pursuers is to learn a cooperative policy that ensures: $\exists i \in \mathcal{P}, \|p_i - p_{tar}\| < R_{cap}$, where R_{cap} is the target capture distance. Simultaneously, the following constraints must be satisfied:

- Collision avoidance among pursuers: $\|p_i - p_j\| > R_s, \forall i, j \in \mathcal{P}, i \neq j$, where R_s is the minimum safety distance.
- Obstacle avoidance: $\|p_i - p_j^{obs}\| > r_j^{obs} + R_s, \forall i \in \mathcal{P}, \forall j \in \mathcal{O}$.

- Kinematic constraints: $|v_i| \leq v_{max}, |u_i| \leq u_{max}, |\omega_i| \leq \omega_{max}, \forall i \in \mathcal{P}$, where u_i and ω_i are the acceleration and the angular velocity of pursuer i respectively.
- Time constraint: $t \in [0, t_{cap}], 0 \leq t_{cap} \leq t_{max}$, where t_{cap} is the time when the target is captured.

These constraints introduce a challenging pursuit task scenario, requiring pursuers to cooperate effectively despite their limited perception capabilities and the target's superior maneuverability.

B. Dec-POMDP of Cooperative Pursuit-Evasion

1) *State Representation*: Let $\xi_{tar} := (x_{tar}, y_{tar}, v_{tar}^x, v_{tar}^y)$ be the target's state, $\xi_i := (x_i, y_i, v_i^x, v_i^y)$ represent pursuer i 's state and $\xi_{obs}^j := (x_j^{obs}, y_j^{obs}, r_j^{obs})$ describe obstacle j 's state, where (x, y) is the planar position, v^x and v^y are velocity components. The global system state can be represented as $s := (\xi_{tar}, \xi_{pur}, \xi_{obs})$, where $\xi_{pur} := \{\xi_i \mid i \in \mathcal{P}\}$ and $\xi_{obs} := \{\xi_j \mid j \in \mathcal{O}\}$ denote the concatenation of all pursuers' states and all obstacles' states respectively.

2) *Observation Space*: From the perspective of pursuer i , its observation can be denoted as $o_i = (\xi_i^{tar}, \xi_i^{self}, \xi_i^{nei}, \xi_i^{obs})$, where $\xi_i^{tar} := (x_{tar} - x_i, y_{tar} - y_i, v_{tar}^x, v_{tar}^y)$ represents the relative planar position and velocity components of the target with respect to pursuer i ; $\xi_i^{self} := (v_i^x, v_i^y, \phi_i, v_i)$ denotes the pursuer's own velocity components, heading angle, and speed; $\xi_i^{obs} := \big\|_{k=1}^{n(d_{i,k}^{obs} < R_c)} (x_k^{obs} - x_i, y_k^{obs} - y_i, r_k, d_{i,k}^{obs})$ contains information on the n nearest obstacles, $\xi_i^{nei} := \big\|_{j=1}^{m(d_{i,j}^{nei} < R_c)} (x_j^{nei} - x_i, y_j^{nei} - y_i, \phi_j^{nei}, v_j^{nei})$ encapsulates information about the m nearest neighboring pursuers. Here, $\|$ is the concatenation operation. To account for the time-varying number of entities within the sensing range R_c , if fewer than m pursuers or n obstacles are detected, the remaining slots in ξ_i^{nei} and ξ_i^{obs} will be filled with zero vectors respectively.

3) *Action Space*: Pursuers' motion is controlled by modifying its linear acceleration and angular velocity. For pursuer i , we define the action space as $a_i := (u_i, \omega_i)$, where $u_i \in (-5, 5)$ m/s² represents the linear acceleration and $\omega_i \in (-\pi/4, \pi/4)$ denotes the angular velocity. These actions are applied in continuous space.

Let v_i and ϕ_i denote the current velocity and heading angle of pursuer i respectively. Then the next velocity setpoint v_i^{ref} and heading angle setpoint ϕ_i^{ref} can be determined by the following equations:

$$\begin{cases} v_i^{\text{ref}} = \text{clip}(v_i + u_i \Delta t, 0, v_{\max}) \\ \phi_i^{\text{ref}} = \arctan2(\sin(\phi_i + \omega_i \Delta t), \cos(\phi_i + \omega_i \Delta t)) \end{cases}, \quad (1)$$

where Δt is the time step. Note that v_i^{ref} is clipped to $[-v_{\max}, v_{\max}]$ and ϕ_i^{ref} is automatically wrapped to $[-\pi, \pi)$.

4) *State Transition Model*: This study adopts the unicycle kinematic model presented in [27], [28] as the foundation for agent motion dynamics. We employ this model to generate simulation data for pursuit-evasion scenarios, which serves as the basis for training our DRL-based cooperative pursuit policy.

5) *Reward Function*: To promote cooperation among pursuers, we implement a team-based reward strategy. This strategy first calculates individual pursuer rewards, then combines and averages them to create a unified team reward. For pursuer i , we define the individual reward r_i as follows:

$$r_i = r_i^{\text{cap}} + r_i^{\text{nei}} + r_i^{\text{obs}}, \quad (2)$$

where r_i^{cap} denotes the *target capture reward*, r_i^{nei} represents the *neighbor avoidance reward*, and r_i^{obs} is the *obstacle avoidance reward*. Specifically, to encourage pursuers to capture the target, the *capture reward* is defined as

$$r_i^{\text{cap}} = \begin{cases} 1000 & \rho_i < R_{\text{cap}} \\ -\rho_i & \rho_i \geq R_{\text{cap}} \end{cases}, \quad (3)$$

where ρ_i is the distance between pursuer i and the target.

To prevent collisions among pursuers, the *neighbor avoidance reward* is designed as

$$r_i^{\text{nei}} = \sum_{j \in \mathcal{N}_i} \begin{cases} -1000 & D_{ij} < R_s \\ 5(D_{ij} - R_a) & R_s \leq D_{ij} \leq R_a \\ 0 & D_{ij} > R_a \end{cases}, \quad (4)$$

where D_{ij} is the distance between pursuer i and pursuer j , $\mathcal{N}_i$ is the set of all other pursuers, and R_a is the alert radius.

Additionally, to avoid collisions with obstacles, the *obstacle avoidance reward* is defined as

$$r_i^{\text{obs}} = -1000 \cdot \sum_{k \in \mathcal{O}} \mathbb{I}(D_{ik}^{\text{obs}} < R_s + r_k^{\text{obs}}), \quad (5)$$

where D_{ik}^{obs} is the distance between pursuer i and obstacle k and $\mathbb{I}(\cdot)$ is the indicator function.

Finally, the team final reward for each pursuer r_{team} is calculated as the mean of all pursuers' individual rewards:

$$r_{\text{team}} = \frac{1}{N} \sum_{i=1}^N r_i. \quad (6)$$

Using the mean as the team reward keeps the signal size-invariant and treats every pursuers contribution equally. This size-independent, low-variance feedback stabilizes gradients

and promotes faster cooperative learning. Importantly, although individual reward components r_i are calculated for each pursuer, the final reward used for learning and decision-making is the team reward r_{team} . This approach ensures a shared objective among all team members.

III. METHOD

In this section, we introduce the knowledge enhanced multi-agent twin delayed deep deterministic policy gradient (KE-MATD3) algorithm for learning effective team strategies in complex pursuit-evasion scenarios. Building on the improved artificial potential field (IAPF) method, we embed domain-specific heuristics into KE-MATD3 through a dedicated knowledge enhancement mechanism, as illustrated in Fig. 1(b). This design leverages a shared team reward to foster cooperative behaviors and tackle challenges such as credit assignment and sparse feedback commonly encountered in multi-agent environments.

A. IAPF Method

Traditional artificial potential field (APF) [29] methods often assume global environmental awareness, neglecting real-world sensing limitations. In this paper, we propose an improved APF (IAPF) method that incorporates sensing range constraints to guide pursuers in collectively capturing a moving target. For pursuer i , IAPF calculates the total force $\mathbf{F}_i$ as

$$\mathbf{F}_i = \mathbf{F}_i^{\text{att}} + \mathbf{F}_i^{\text{nei}} + \mathbf{F}_i^{\text{obs}}, \quad (7)$$

where $\mathbf{F}_i^{\text{att}}$, $\mathbf{F}_i^{\text{nei}}$, and $\mathbf{F}_i^{\text{obs}}$ represent the *target attraction force*, *inter-pursuer repulsion force*, and *obstacle repulsion force* respectively. In this formulation, $\mathbf{F}_i^{\text{att}}$ is defined as a unit vector pointing towards the target. For pursuer i , $\mathbf{F}_i^{\text{nei}}$ is designed to prevent collisions between pursuers within the sensing range R_c :

$$\mathbf{F}_i^{\text{nei}} = \sum_{j \in \mathcal{N}_i} \begin{cases} \eta_0 \left(\frac{1}{D_{ij}} - \frac{1}{d_0} \right) \frac{1}{D_{ij}^2} \mathbf{n}_{ij}, & D_{ij} < d_0 \\ 0, & \text{otherwise} \end{cases}. \quad (8)$$

Here, η_0 is the inter-pursuer repulsion coefficient, d_0 is the repulsion range ($d_0 < R_c$), and $\mathbf{n}_{ij}$ is the unit vector pointing from pursuer j to pursuer i . $\mathbf{F}_i^{\text{nei}}$ creates a repulsive force that intensifies as pursuers approach each other, peaking at close range and gradually diminishing to zero at the repulsion boundary. Similarly, the *obstacle repulsion force* $\mathbf{F}_i^{\text{obs}}$ is defined as

$$\mathbf{F}_i^{\text{obs}} = \sum_{k \in \mathcal{O}} \begin{cases} \eta_1 \left(\frac{1}{\rho_{ik}} - \frac{1}{d_0} \right) \frac{1}{\rho_{ik}^2} \mathbf{n}_{ik}, & \rho_{ik} < d_0 \\ 0, & \text{otherwise} \end{cases}, \quad (9)$$

where η_1 is the obstacle repulsion coefficient, $\rho_{ik} := D_{ik}^{\text{obs}} - r_k^{\text{obs}}$ is the distance between pursuer i and the edge of obstacle k and $\mathbf{n}_{ik}$ is the unit vector pointing from obstacle k to pursuer i . This force facilitates effective collision avoidance while allowing smooth navigation near obstacles.

B. Knowledge-Enhanced MATD3

To address the challenge of learning from team rewards, we propose a knowledge-enhanced reinforcement learning module based on the MATD3 algorithm (KE-MATD3). This module integrates heuristic knowledge from the IAPF method into MATD3, synergizing domain-specific expertise with data-driven learning.

Specifically, MATD3 extends the twin delayed deep deterministic policy gradient (TD3) [30] algorithm to multi-agent settings. In MATD3, each agent maintains a pair of critic networks $\{Q_1, Q_2\}$. These critics are updated by minimizing the following loss:

$$\mathcal{L}_{Q_i}(\theta^{Q_i}) = \mathbb{E}_{\tau \sim \mathcal{D}} \left[(Q_i(s_j, \mathbf{a}_j | \theta^{Q_i}) - y_j)^2 \right], i \in \{1, 2\}, \quad (10)$$

where τ is an experience sample from the replay buffer $\mathcal{D}$, with j denoting the index of the transition. The target value y_j is computed as

$$y_j = r_{\text{team}} + \gamma \min_{i=1,2} \left[Q_i^-(s'_j, \tilde{\mathbf{a}}' | \theta^{Q_i^-}) |_{\tilde{\mathbf{a}}'_j = \mu^-(o_j | \theta^{\mu^-})} \right], \quad (11)$$

where Q_i^- and μ^- denote slowly updated target critic and target actor respectively. The actor (policy) network is updated less frequently (once every C critic updates) with the following objective:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{\tau \sim \mathcal{D}} \left[Q_1(s_j, \tilde{\mathbf{a}} |_{\tilde{\mathbf{a}}'_j = \mu^-(o_j | \theta^{\mu^-})}) \right]. \quad (12)$$

By delaying the actor updates, MATD3 obtains more reliable target estimates from its critics, thereby reducing oscillations and improving overall training stability.

To seamlessly integrate the IAPF velocity constraint into our knowledge-consistency objective, we define a unified alignment loss that enforces both velocity feasibility and high-level heuristic consistency. Specifically, we introduce a projection operator $\Pi(\mathbf{v})$ that maps any velocity vector $\mathbf{v}$ into the closed ball of radius v_{max} . Formally,

$$\Pi(\mathbf{v}) = \mathbf{v} \min \left(1, \frac{v_{\text{max}}}{\|\mathbf{v}\|} \right). \quad (13)$$

Under IAPF, the desired velocity of pursuer i is given by $\Pi(\mathbf{F}_i)$, where $\mathbf{F}_i$ is the IAPF-derived force. Meanwhile, as illustrated in Fig. 1(c), the actions output by the policy network are converted into the final velocity v_i^{RL} using Equation (1). To align the learned behavior with this heuristic feedback, we define the knowledge-consistency loss $\mathcal{L}_{\text{KC}}$:

$$\mathcal{L}_{\text{KC}} = \mathbb{E} \left[\|\Pi(\mathbf{F}_i) - v_i^{\text{RL}}\|^2 \right]. \quad (14)$$

Minimizing $\mathcal{L}_{\text{KC}}$ promotes the alignment of the learned behavior with the IAPF-driven velocity constraint, effectively reducing exploration blindness and enhancing learning efficiency. In contrast to traditional imitation learning, this formulation positions IAPF's velocity feedback as soft alignment rather than a strict teacher signal, thereby preserving the core exploration mechanisms of reinforcement learning.

Notably, KE-MATD3 does not directly convert the IAPF velocity output $\Pi(\mathbf{F}_i)$ into corresponding acceleration and

Algorithm 1: KE-MATD3

Input: maximum training episode N_{max} , replay buffer maximum capacity N_{buffer} , maximum time step per episode t_{max} , training batch size N_b , update frequency of target networks C , replay buffer $\mathcal{D}$, policy network parameters θ^μ , critic network parameters $\theta^{Q_1}, \theta^{Q_2}$, target network parameters $\theta^{\mu^-}, \theta^{Q_1^-}, \theta^{Q_2^-}$;
Output: policy network parameters θ^μ ;
1 Initialize actor network μ and critic network Q_1, Q_2 with parameters θ^μ, θ^{Q_1} and θ^{Q_2} randomly;
2 $\theta^{\mu^-} \leftarrow \theta^\mu, \theta^{Q_1^-} \leftarrow \theta^{Q_1}, \theta^{Q_2^-} \leftarrow \theta^{Q_2}$;
3 **for** episode $e = 1$ to N_{max} **do**
4 Reset the task environment and receive initial observations $\mathbf{o}$ and states $\mathbf{s}$;
5 **for** $t \leftarrow 1$ to N_s **do**
6 Create observation $\mathbf{o}_i$ for each pursuer i ;
7 Select and execute action $\mathbf{a}_i = \mu(\mathbf{o}_i | \theta^\mu) + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2, -c, c)$;
8 Calculate team reward r_{team} with (6);
9 Receive new observations $\mathbf{o}'$ and states $\mathbf{s}'$;
10 Store $(\mathbf{s}, \mathbf{o}, \mathbf{a}, r, \mathbf{o}', \mathbf{s}')$ in replay buffer $\mathcal{D}$;
11 $\mathbf{s} \leftarrow \mathbf{s}', \mathbf{o} \leftarrow \mathbf{o}'$;
12 Sample a minibatch of N_b tuples from $\mathcal{D}$;
13 Update θ^{Q_i} by minimizing (10);
14 **if** $t \equiv 0 \pmod{C}$ **then**
15 Update θ^μ by minimizing (15);
16 Update target network parameters θ^{μ^-} and $\theta^{Q_i^-}$ with soft update rate τ :
17 $\theta^{\mu^-} \leftarrow \tau \theta^\mu + (1 - \tau) \theta^{\mu^-}$,
18 $\theta^{Q_i^-} \leftarrow \tau \theta^{Q_i} + (1 - \tau) \theta^{Q_i^-}, i = 1, 2$;
19 **end**
20 **if** $\exists i \in \mathcal{P}, \|p_i - p_{\text{tar}}\| < R_{\text{cap}}$ **then**
21 Reset the task environment and receive new initial observations $\mathbf{o}$ and states $\mathbf{s}$;
22 **end**
23 **end**

angular velocity to construct $\mathcal{L}_{\text{KC}}$ with the MATD3 output. This choice stems from the limitations of IAPF: although $|\Pi(\mathbf{F}_i)| \leq v_{\text{max}}$ is guaranteed, there is no assurance that the resulting acceleration $\mathbf{u}$ and angular velocity ω satisfy lower-level kinematic constraints. In other words, IAPF serves as a suboptimal guiding policy, providing valuable velocity heuristics but unable to fully optimize the team reward. Consequently, the primary exploration still relies on reinforcement learning. To balance IAPF's heuristic guidance with the breadth of RL exploration, the total policy-network loss $\mathcal{L}_{\text{Total}}$ is given by

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{RL}} + \lambda \mathcal{L}_{\text{KC}}, \quad (15)$$

where λ is a weighting coefficient that adjusts the influence of IAPF heuristics in shaping the learning process. By integrating partial domain knowledge through $\mathcal{L}_{\text{KC}}$, the

approach achieves a synergy between model-based guidance and reinforcement learning, thereby enhancing performance and learning stability.

To further enhance learning efficiency, we employ parameter sharing mechanism [31] across agents. Algorithm 1 summarizes KE-MATD3. It is worth noting that while we employ MATD3 as the backbone algorithm in this work, the proposed knowledge enhancement module is generic and can be integrated with any MADRL algorithm.

IV. EXPERIMENTS AND RESULTS

In this section, we evaluate our KE-MATD3 algorithm in numerical simulation and real-world experiment.

A. Parameters Settings

We implemented our algorithm using the PyTorch library. As illustrated in Fig. 1(b), the policy network employed a three-layer MLP architecture with 256, 256, and 2 neurons in each layer respectively. For the critic component, we utilized two separate value networks, each consisting of a four-layer MLP with 256, 128, 64, and 1 neurons in successive layers. All network parameters were updated using the Adam optimizer, with learning rates of 5×10^{-4} for the actor and 1×10^{-3} for each critic. During training, the exploration parameter σ was exponentially annealed from 1.0 to 0.05, and the weighting coefficient λ was set to 1 to balance learning and knowledge enhancement.

The proposed algorithm was trained for 3×10^5 steps in a 200×200 m square environment with $N = 4$ pursuers and $M = 20$ obstacles. The radii of obstacles were randomly generated values ranging from 5 m to 10 m. Upon target capture or mission timeout, the environment was reset, randomizing the initial positions of the target and pursuers, as well as the locations and sizes of obstacles. We conducted 5 independent training runs with different random seeds for all experiments. Table I presents the empirical values for other essential parameters.

TABLE I
PARAMETER SETTINGS

Name	Value	Name	Value
R_{cap}	5 m	R_c	30 m
R_a	20 m	R_s	1 m
v_{max}	5 m/s	u_{max}	5 m/s ²
m	2	n	2
η_0	500	η_1	800
N_{max}	3000	t_{max}	100
N_{buffer}	1×10^5	N_b	32
τ	0.01	γ	0.95

B. Numerical Simulation Results

We first compared our KE-MATD3 algorithm with several baselines in numerical simulation.

1) *Training Results*: To ensure a comprehensive evaluation, we selected several benchmark algorithms: variants of vanilla MATD3 [23], MADDPG [26], MADDQN [32] with different reward structures, and the IAPF. Specifically,

we denoted algorithms learning directly from the team reward r_{team} without using KE mechanism as T-MATD3, T-MADDPG, and T-MADDQN. Their counterparts using the individual reward r_i were termed I-MATD3, I-MADDPG, and I-MADDQN respectively. To maintain fairness in our comparisons, all learning-based algorithms utilized same network architectures and hyperparameters. For the IAPF algorithm, we first converted its output v^{des} into desired acceleration u and angular velocity ω , then applied value clipping to the intervals $[-u_{max}, u_{max}]$ and $[-\omega_{max}, \omega_{max}]$ respectively to comply with the kinematic constraints.

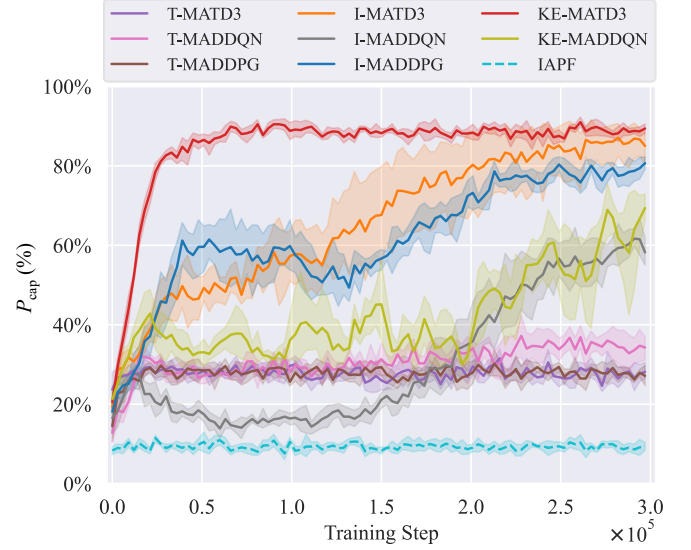


Fig. 2. Evolution of the average capture success rate P_{cap} for different algorithms during training. The shaded region represent the standard error.

We evaluated algorithmic learning performance by conducting tests every 100 training steps, with each test comprising 10 episodes. Fig. 2 illustrates the evolution of average capture success rate P_{cap} for all algorithms. Successful capture, as defined in Section II-A, requires target apprehension within the specified time without any collision of each pursuer. Noting that IAPF which is a non-learning method exhibits performance variance solely due to environmental stochasticity in tests. As evident from Fig. 2, T-MATD3, T-MADDPG, T-MADDQN show minimal P_{cap} improvement during training. In contrast, algorithms utilizing individual rewards (i.e., I-MATD3, I-MADDPG and I-MADDQN) demonstrate significant upward trends, indicating greater tractability. Notably, KE-MATD3 and KE-MADDQN, which utilize the knowledge enhancement (KE) mechanism with team rewards, outperform their individual reward counterparts (i.e., I-MATD3 and I-MADDQN) in final P_{cap} . Despite IAPF's low P_{cap} ($\sim 10\%$), the KE mechanism still effectively extracts useful information, steering the learning process toward higher-quality policies. Furthermore, compared to KE-MADDQN, KE-MATD3 converges faster due to the advantage of algorithm architecture. It achieves the highest final P_{cap} with minimal variance, demonstrating its superior efficiency and performance across all benchmarks.

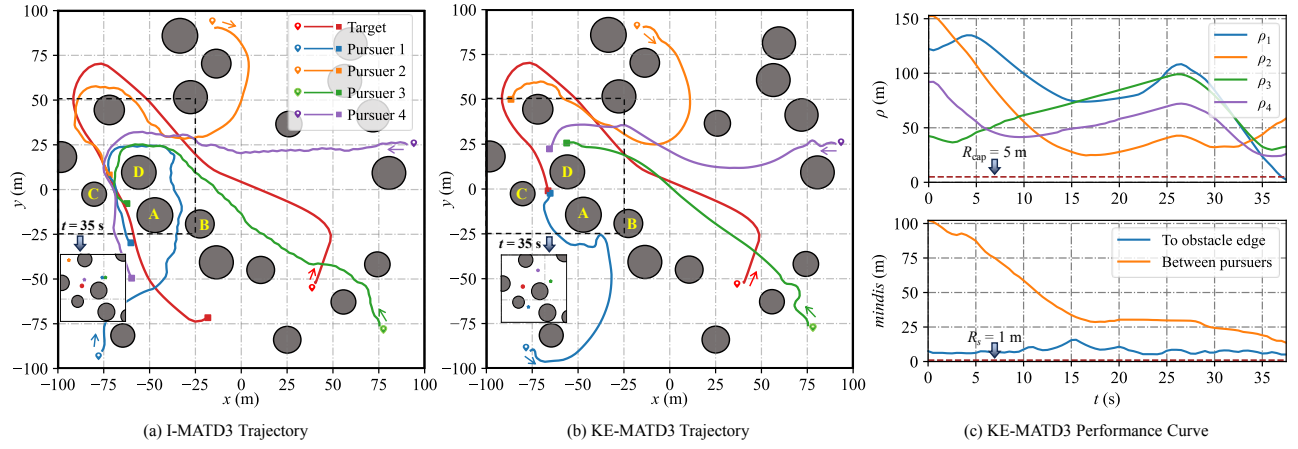


Fig. 3. Visualization of test results under the typical scenario.

TABLE II

QUANTITATIVE TESTING RESULTS OF FOUR POLICIES IN PURSUIT-EVASION TASKS WITH DIFFERENT OBSTACLE DENSITY

Method	$M = 20$				$M = 40$				$M = 50$				$M = 60$			
	$P_{cap}(\%)$	$P_{obs}(\%)$	$P_{nei}(\%)$	Score	$P_{cap}(\%)$	$P_{obs}(\%)$	$P_{nei}(\%)$	Score	$P_{cap}(\%)$	$P_{obs}(\%)$	$P_{nei}(\%)$	Score	$P_{cap}(\%)$	$P_{obs}(\%)$	$P_{nei}(\%)$	Score
I-MADDPG	81.80	12.36	1.92	6.65	72.62	22.42	2.20	4.66	67.68	26.24	2.98	3.59	62.12	31.56	2.76	2.53
I-MATD3	84.84	9.46	1.52	7.35	76.28	17.86	2.04	5.50	71.38	21.80	2.26	4.57	66.10	25.58	2.36	3.60
KE-MADDPG	88.68	1.60	0.58	8.72	88.62	6.12	1.04	8.18	85.12	9.12	1.88	7.35	80.90	12.04	2.32	6.51
KE-MATD3	89.92	1.48	0.42	8.90	89.54	4.78	1.18	8.34	85.36	8.94	1.42	7.45	80.78	11.88	2.16	6.53

2) *Qualitative Evaluation*: After training, we tested KE-MATD3 in a typical pursuit-evasion task. The agents' trajectories are illustrated in Fig. 3(b). Initially, the target moved towards the upper right, while all pursuers adeptly avoided their nearest obstacles and maneuvered towards it. When the target adjusted its heading to the upper left corner, pursuer 2, 3, and 4 swiftly adapted their trajectories, collectively forcing the target into a confined space. Notably, pursuer 1 strategically circumnavigated obstacle A from below, effectively coordinating to outflank the target. At $t = 35$ s, all pursuers established a sophisticated containment formation: pursuer 3 and 4 blocked the target's escape route to the upper right, pursuer 2 obstructed its path to the upper left, while pursuer 4 positioned itself below for interception. At $t = 38$ s, the target was captured by pursuer 1. Fig. 3(c) shows the performance curves of KE-MATD3, illustrating no collisions occurred throughout the entire pursuit process. These results highlight KE-MATD3's ability to foster co-operative behaviors for target encirclement and capture in complex environments.

We then visualized I-MATD3 trajectories under the same initial conditions in Fig. 3(a). While I-MATD3 initially behaved similarly to KE-MATD3, a significant divergence occurred when pursuer 1 encountered obstacles A and B. In contrast to KE-MATD3, pursuer 1 of I-MATD3 attempted to pass through the narrow gap in pursuit of proximity to the target. This decision not only elevated the collision risk with pursuer 3 but also failed to establish an effective encirclement formation. At $t = 35$ s, all pursuers were positioned above the target, leaving the lower escape route unguarded. As a result, the target successfully escaped through the lower right

corner formed by obstacle C and D. This outcome highlights I-MATD3's tendency to prioritize short-term gains over long-term benefits, resulting in locally optimal solutions.

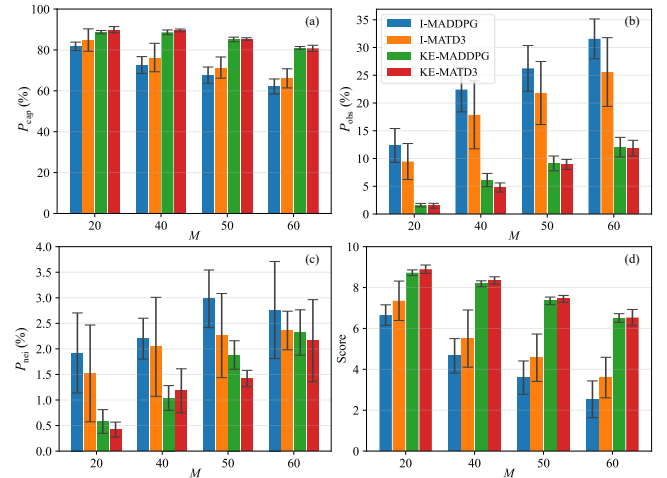


Fig. 4. Bar plot for testing results of the four learned policies. Note that the error bar is used to show the standard error. (a) Average capture success rate. (b) Average pursuer-obstacle collision rate. (c) Average inter-pursuer collision rate. (d) Average task score.

3) *Quantitative Evaluation*: To rigorously measure generalization performance, we tested KE-MATD3, KE-MADDPG, I-MATD3, and I-MADDPG with different number of obstacles. Table II and Fig. 4 presents a comparison of their performance by averaging the results from 1000 experiments. In Table II, M denotes the number of obstacles; P_{cap} the average capture success rate; P_{obs} the average

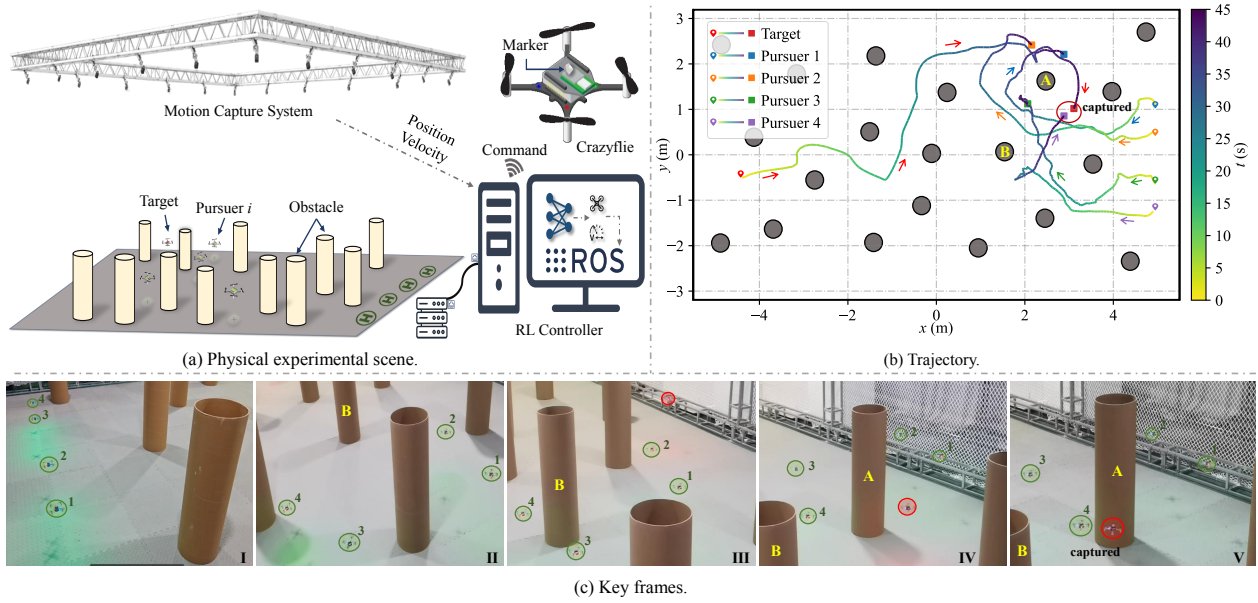


Fig. 5. Physical experimental Results. The flight video is provided in the supplementary material.

pursuer-obstacle collision rate; P_{nei} the average inter-pursuer collision rate. We introduced a ‘score’ metric to evaluate average task performance, assigning +10 for successful target capture and -10 for any collision event. This scoring system was preferred over raw rewards to mitigate potential misinterpretations arising from scenarios where collision-induced early terminations might yield higher rewards than successful but prolonged pursuits, due to reduced accumulation of the negative per-time-step reward component ρ . Intuitively, algorithms incorporating the KE mechanism demonstrated enhanced performance across all metrics. KE-MATD3, in particular, consistently outperformed others in nearly every aspect. This means our proposed algorithm possesses robust zero-shot generalization capabilities, maintaining its efficacy across environments with different obstacle densities.

C. Real-World Experiment Results

We finally carried out a real-world experiment to evaluate the practicability of the learned KE-MATD3 policy.

1) *Physical Experimental Setup*: As illustrated in Fig. 5(a), the real-world system comprises 5 Crazyfly 2.1 [33] quadcopters, a NOKOV motion capture system, 20 cylindrical obstacles, and an onboard computer. Each quadcopter, measuring $92 \times 92 \times 29$ mm, serves as an agent in our experiments and is equipped with an LED module that displays green for pursuers and red for the target. The motion capture system tracks 16 mm-diameter markers on the Crazyflies, determines their real-time positions and velocities, and relays this data to the onboard computer via ROS messages. The experimental arena measures $6.4 \times 11 \times 2$ m and contains cylindrical obstacles with a 20 cm radius and 1 m height. All quadcopters maintain a fixed altitude of 0.5 m throughout the experiment. The onboard computer executes the trained policy network and transmits control commands wirelessly to the quadcopters at 400 Hz. Due to the arena constraints,

the targets is capped at 0.25 m/s, while pursuers are limited to $v_{\max} = 0.25/1.5 = 0.167$ m/s.

2) *Trajectory and Key Frame Analysis*: Fig. 5(b) depicts the motion trajectories of pursuers and the target, with color intensity indicating time progression. At the beginning, all pursuers took off from the right side of the arena and ascended to 0.5m altitude (Fig. 5(c)-I). Then, they moved towards the target in a dispersed formation to minimize the risk of inter-agent collisions (Fig. 5(c)-II). At $t = 27$ s, a tactical divergence occurred: pursuers 1, 2, and 3 continued direct pursuit, while pursuer 4 hovered near obstacle B in an ambush position (Fig. 5(c)-III). This strategy proved effective at $t = 43$ s, when pursuers 1, 2, and 3 forced the target to circle around obstacle A into the lower left quadrant. Seizing the opportunity, pursuer 4 immediately sprang into action, coordinating with the others to intercept the target near obstacle A (Fig. 5(c)-IV). Ultimately, at $t = 45$ s, the target was captured by pursuer 4 (Fig. 5(c)-V). This result demonstrates that the learned KE-MATD3 policy is capable of safely and efficiently completing the capture task while enabling emergent cooperative behaviors among pursuers.

V. CONCLUSION

This paper introduces KE-MATD3, a novel team reward based knowledge-enhanced cooperative pursuit algorithm tailored for capture tasks in cluttered environments. By integrating an improved artificial potential field (IAPF) method with a knowledge enhancement (KE) mechanism into the MATD3 algorithm, we effectively leverage domain-specific heuristics to guide policy learning. Although IAPF alone may not fully account for all kinematic constraints, the KE mechanism extracts valuable information from the method, thereby expediting learning. Numerical simulations illustrate KE-MATD3s superior learning efficiency and performance compared to various baselines, while approaches that incorporate the KE

mechanism consistently outperform individual reward based counterparts. Real-world experiments further confirm these findings, revealing coordinated behaviors among pursuers. Looking ahead, this algorithm could be extended to three-dimensional, multi-target cooperative capture scenarios in the future.

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